

Generalized Gaussian Distribution

I. INTRODUCTION

As today's communication scenarios are becoming ever more complex, the classical assumptions of Gaussian noise can no longer capture the multitude of possible communication scenarios. The goal of this work is to study a large family of probability distributions termed Generalized Gaussian (GG) that started to receive attention in many important communication scenarios. We shall refer to X with GG distribution given by the probability density function (pdf)

$$f_X(x) = \frac{c_p}{\alpha} e^{-\frac{|x-\mu|^p}{2\alpha^p}}, c_p = \frac{p}{2^{\frac{p+1}{p}} \Gamma\left(\frac{1}{p}\right)}, x \in \mathbb{R}, \quad (1)$$

as $X \sim \mathcal{N}_p(\mu, \alpha^p)$.¹ Another commonly used name, especially in economics, is Generalized Error Distribution. The flexible parametric form of the pdf of GG distribution allows for tails that are either heavier² than Gaussian ($p < 2$) or lighter than Gaussian for ($p > 2$) which makes it an excellent choice for many modeling scenarios. The origin of the GG family can be trace to the seminal work of Levy [1]. The famous cases of this distribution include:

- Laplace distribution for $p = 1$;
- Gaussian distribution for $p = 2$; and
- Uniform distribution on $[-\beta, \beta]$ for $p = \infty$ and $\alpha = \lim_{p \rightarrow \infty} \left(\frac{1}{2}\right)^{\frac{1}{p}} \beta$.

CDF DISTRIBUTION We also know the expression for the CDF of X which is given by

$$F_X(x) = \frac{1}{2} + \text{sign}(x - \mu) \frac{\gamma\left(\frac{1}{p}, \frac{|x-\mu|^p}{2\alpha^p}\right)}{2\Gamma\left(\frac{1}{p}\right)}, x \in \mathbb{R}. \quad (2)$$

A. Past Work

In communication theory, the GG distribution finds many modeling applications in the so called *impulsive noise scenarios* which occur when the noise pdf has a greater tail area than the Gaussian pdf, i.e. heavier than Gaussian. For example, in [2] it is shown that in ultrawideband (UWB) systems with time-hopping (TH) the interference should be modeled with probability distributions that are more impulsive than Gaussian. Moreover, it has been shown that for the moderate and high signal-to-noise ratio (SNR) the TH-UWB the interference is well modeled by the GG distribution with parameter $p \leq 1$. In [3], [4] many atmospheric noises were shown to be very impulsive and values of $0.1 < p < 0.6$ were shown to be good approximations.

The GG distribution can also model noise that appear in non-standard wireless mediums. For example, in [5] the GG has been recognized as a model for the underwater acoustic channel where values of $p = 2.2$ and $p = 1.6$ have been found to model ship transit noise and sea surface agitation noise, respectively. Moreover, authors of [5] also studied error probability of classical communication constellations in the GG noise with the traditional AWGN receiver and found that the performance significantly degrades when the noise kurtosis is greater than the Gaussian kurtosis (i.e., fourth standardized moment). In [6] the GG distribution has been shown to model RF signals in medical applications that are involved in the recovery of echocardiogram images.

Other application of the GG distribution include modeling of: Gabor coefficients in face-recognition problems [7]; load demand in power systems [8]; pixels forming fine-resolution synthetic aperture radar (SAR) images [9]; and texture retrieval problems [10].

Because the pdf of GG distribution has a very simple form, many quantities such as moments, entropy and Reyni entropy can be easily computed [10], [11]. The log likelihood function can also be computed which make it possible to study problems of designing optimal detectors for signals in the presence of the GG noise see [4], [12] and the reference therein.

From the information theoretic perspective the GG distribution is also interesting because it maximizes the entropy and Reyni entropy under the p -th absolute moment constraints [13], [14]. The maximum entropy property can serve as an important intermediate step in a number of proofs. For example, in [15] it has been used to generalize the so-called Ozarow-Wyner bound [16] on the mutual information of discrete inputs over arbitrary channels. Also, in [17] the maximum entropy principle has been used to improve bounds on the entropy of Gaussian mixtures.

While the number of applications of the GG distribution is vast, regrettfully many of the result are drawn from the numerical result, and little is know about the GG family beyond $p = 1, 2$ and $p = \infty$. For instance, very little is know about the characteristic function of the GG distribution and only expressions in terms of hypergeometric functions are known. For example, the characteristic function of the GG distribution was given in terms of Fox-Write functions in [18] for all $p > 1$

¹We define the Gamma function as $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$.

²If the tail area of the pdf is greater (smaller) than the tail area of the Gaussian pdf, then we say that the pdf has a heavier (lighter) tail than Gaussian.

and later generalized in terms of Fox-H functions in [19] to $p > 0$. The work of [19] also characterized the pdf of the sum of two independent GG random variables in terms Fox-H function. Some non-linear transformations of sums of independent GG distributions and the moment generating function of the GG distribution have been studied in [20].

B. Paper Outline and Contributions

Our contributions are as follows:

- 1) In Section III, Theorem 1 connects the pdf of GG distributions to *positive definite* functions. In particular, we show that for $p \leq 2$ the pdf of the GG distribution is a positive definite function and for $p > 2$ the pdf is not a positive definite function. Moreover, it is shown that for $p \leq 2$ the pdf of the GG distribution can be expressed as an integral of a Gaussian pdf with respect to some non-negative finite Borel measure.
- 2) In Section II, derives inequalities and properties of moments and absolute moments of the GG distribution.
- 3) In Section IV, we study properties of the characteristic function of the GG distribution and show:
 - In Section IV-A, Proposition 4 shows under what conditions the characteristic function of the GG distribution is a real analytic function;
 - In Section IV-B, Theorem 3 studies the distribution of zeros of the characteristic function of the GG distribution. In particular, it is shown that for $p \leq 2$ the characteristic function of the GG distribution has no zeros and is always positive and for $p > 2$ the characteristic function has at least one positive-to-negative zero crossing; and
 - In Section IV-C, we study the conditions under which a GG distribution of order p can be additively transformed into another GG distribution of order q . In particular, Theorem 4 gives the conditions on (p, q) under which such transformation is not possible. Moreover, Proposition 6 gives a sufficient conditions on the characteristic function to guarantee that such a transformation exists.
- 4) In Section V we study properties of information measures of the GG distribution and show:
 - In Section V-A, we review properties of the differential entropy of the GG distribution;
 - In Section V-B, Proposition 8 studies regularity properties of the GG distribution;
 - In Section V-C, Proposition 9 show the derivative of the mutual information with respect to signal-to-noise ratio (SNR) of an additive channel channel with the GG distributed noise; and
 - In Section V-D, Proposition 12 some properties of the relative entropy of the GG distribution are considered.
- 5) In Section VI we study properties of an additive channel channel with the GG distributed noise of order p under the q -th moment constraint and show:
 - In Section VI-A studies the conditions under which the Shannon's upper bound on the capacity in Proposition 13 is achievable. Besides the well known case of $(p, q) = (2, 2)$ (i.e., Gaussian noise and the second moment constraint), in Proposition 14 it is shown that the Shannon's upper bound is also tight for $(p, q) = (1, 1)$ (i.e., Laplace noise and the absolute first moment constraint). However, in Proposition 15, using the deconvolution results from Section IV-C, it is shown that in general the Shannon's upper bound is not achievable. In Proposition 16, it is shown that for any (p, q) the Shannon's upper bounded is achievable with in a constant gap by a uniformly spaced equally distributed discrete input, implying that discrete inputs achieves a full degree of freedom;
 - In Section VI-B, Proposition 17 it is shown that the GG distributed input can also be optimal with in an additive gap;
 - In Section VI-C, Proposition 19 it is shown that when p and q are even integers than the input distribution is discrete with finitely many points; and
 - In Section VI-D, Proposition 20 we study the case of $p = \infty$ (amplitude constraint) and give a condition under which the optimal input distribution is binary (i.e., ± 1 equally likely).

C. Notation

TBD

II. MOMENTS

The moments and absolute moments of GG distribution are given next.

Proposition 1. (Moments [11].) *For any $p > 0$ and $k > -1$ the moments of $X \sim \mathcal{N}_p(0, \alpha^p)$ are given by*

$$\mathbb{E}[X^k] = \frac{2^{\frac{k}{p}} \alpha^k}{\Gamma\left(\frac{1}{p}\right)} \Gamma\left(\frac{k+1}{p}\right), k = \text{even}, \quad (3)$$

$$\mathbb{E}[X^k] = 0, k = \text{odd}. \quad (4)$$

Moreover, for any $p > 0$ and $k > -1$ the absolute moments are given by

$$\mathbb{E}[|X|^k] = \frac{2^{\frac{k}{p}} \alpha^k}{\Gamma\left(\frac{1}{p}\right)} \Gamma\left(\frac{k+1}{p}\right). \quad (5)$$

Proof: Moments of the GG distribution can be easily compute by using the integral

$$\int_0^\infty x^k e^{-x^p} dx = \frac{1}{p} \Gamma\left(\frac{k+1}{p}\right), \quad (6)$$

and a change of variable, and where the above integral is finite if $k > -1$ and $p > 0$. ■

Note the p -th absolute moment of $X \sim \mathcal{N}_p(0, \alpha^p)$ is given by

$$\mathbb{E}[|X|^p] = \frac{2\alpha^p}{p}. \quad (7)$$

The following corollary, which relates k -th moments of two GG distributions of a different order, is useful in many proofs.

Corollary 1. Let $Y \sim \mathcal{N}_q(0, 1)$ and $Z \sim \mathcal{N}_p(0, 1)$ then for $q \geq p$

$$\mathbb{E}[|Y|^k] \leq \mathbb{E}[|Z|^k], \quad (8)$$

for any $k \in \mathbb{R}^+$. Moreover, for $q > p$

$$\lim_{k \rightarrow \infty} \left(\frac{\mathbb{E}[|Z|^k]}{\mathbb{E}[|Y|^k]} \right)^{\frac{1}{k}} = \infty. \quad (9)$$

Proof: See Appendix A. ■

The classical moment problem asks whether a distribution can be uniquely determined by its moments. If the the answer is affirmative we say that the moment problem is *determinate*. Otherwise, we say that the problem is *indeterminate* and there might exists another distribution that shares the same moments.

The next result might be somewhat surprising.

Proposition 2. The GG distribution is determinate for $p \geq 1$ and indeterminate for $p < 1$.

Proof: We first show that for $p < 1$ the GG distribution is indeterminate. To show that an absolutely continuous distribution with pdf $f(x)$ is indeterminate one can check a classic *Krein* sufficient condition [1] given by

$$\int_{-\infty}^\infty \frac{-\log(f(x))}{1+x^2} dx < \infty. \quad (10)$$

That is, if (10) is satisfied then the distribution is indeterminate. For the GG distribution the condition in (10) reduces to

$$\int_0^\infty \frac{x^p}{1+x^2} dx, \quad (11)$$

which is finite if $p < 1$. Therefore, for $p < 1$ the GG distribution is indeterminate.

To show that the distribution is determinate it is enough to show that the characteristic function has a power series expansion with a positive radius of convergence. For the GG distribution this will be done in Proposition 4. ■

Give an example of a distribution with the same moments.

Need to find good reference fo Humburger problem.

III. RELATION TO POSITIVE DEFINITE FUNCTIONS

As will be observed throughout this paper the GG distribution will exhibit different properties depending if $p \leq 2$ or $p > 2$. At the heart of this behavior is the concept of positive-definite functions.

Definition 1. A function $f : \mathbb{R} \rightarrow \mathbb{C}$ is called positive definite if for any real numbers $x_1, x_2, \dots, x_n$ the $n \times n$ matrix

$$A = (a_{i,j})_{i,j=1}^n, \quad a_{i,j} = f(x_i - x_j), \quad (12)$$

is positive semi-definit.

Our first result relates the pdf of the GG distribution to the class of positive definite functions.

Theorem 1. The function $e^{-\frac{|x|^p}{2}}$ is

- not positive definite for $p > 2$; and

- positive definite for $0 < p \leq 2$. Moreover, for $x > 0$

$$e^{-\frac{x^p}{2}} = \int_0^\infty e^{-\frac{tx^2}{2}} d\mu(t), \quad (13)$$

where $d\mu(t)$ (independent of x) is some finite non-negative Borel measure on $t \in [0, \infty]$.

Proof. See Appendix B. □

The concept of positive-definite functions will also play an important role in examining properties of the characteristic function of the GG distribution.

A consequence of Theorem 1 we have the following decompositional representation of the GG random variable.

Proposition 3. For any $0 < p \leq 2$ and $X \sim \mathcal{N}_p(0, 1)$ we have that

$$X = V \cdot Z, \quad (14)$$

where V (dependent on p) is a positive random variable independent of $Z \sim \mathcal{N}_2(0, 1)$. Moreover, V has the following properties:

- V is an unbounded random variable for $p < 2$ and $V = 1$ for $p = 2$;
- The power series of the characteristic function of V has a radius of convergence r given by

$$r = \begin{cases} \infty, & p > \frac{2}{3} \\ 0, & p < \frac{2}{3} \end{cases}. \quad (15)$$

Proof: See Appendix C. ■

An interesting future direction is to characterize the distribution of V exactly. This seems possible in light of the fact that all moments of V are known and given by

$$\mathbb{E}[V^k] = \frac{\mathbb{E}[|X|^k]}{\mathbb{E}[|Z|^k]}, k \in \mathbb{R}^+, \quad (16)$$

where $X \sim \mathcal{N}_p(0, 1)$ and $Z \sim \mathcal{N}_2(0, 1)$ and where $\mathbb{E}[|X|^k], \mathbb{E}[|Z|^k]$ are computed in Proposition 1. However, as can be seen from Proposition 3 the characteristic function of V can only be reconstructed from its moments for $p > \frac{2}{3}$. Moreover, it is also not difficult to show that Carleman's sufficient condition for moments to determine the distribution given by

$$\sum_{k=1}^{\infty} (\mathbb{E}[V^k])^{-\frac{1}{k}} = \infty, \quad (17)$$

fails for $p < \frac{2}{3}$ and the series in (17) converges. The fact that for $p < \frac{2}{3}$ the characteristic function of V can not be represented by its power series and that the Carleman's sufficient condition fails seem to suggest that V might not be uniquely determined by its moments for $p < \frac{2}{3}$.

Although, for $p < \frac{2}{3}$ the problem of determining the distribution V might be difficult, for $p > \frac{2}{3}$ we have a complete and unique characterization of the distribution of V in terms of its moments. For example, by examining the moments of V for the case of $p = 1$ one can show that random variable V has a Rayleigh distribution.

Another interesting observation is that while the distribution of V is determinate for $p > \frac{2}{3}$ and Z is always determinate, the distribution of $X = V \cdot Z$ is determinate only for $p \geq 1$ and indeterminate for $p < 1$. This serves as an interesting example, showing that the product of two independent determinate random variables might not be determinate.

Things we would like to learn

- characterization of distribution of V for $p > 2/3$
- is V a countinous random variable

IV. CHARACTERISTIC FUNCTION

The focus of this section is the characteristic function of the GG distribution. This analysis will play an important role in analyzing the output distribution of additive channels with the GG distributed noise. The characteristic function of the GG distribution can be given in the following integral form.

Theorem 2. For any $p > 0$ the characteristic function of $X \sim \mathcal{N}_p(0, \alpha^p)$ is given by

$$\psi_p(t) = 2c_p \int_0^\infty \cos(t\alpha x) e^{-\frac{x^p}{2}} dx. \quad (18)$$

Proof: The proof follows from the fact that $e^{-\frac{|x|^p}{2}}$ is an even function and therefore Fourier transform is equivalent to cosine transform. ■

Examples of characteristic functions of $X \sim \mathcal{N}_p(0, 1)$ for several values of p is given in Fig. 1.

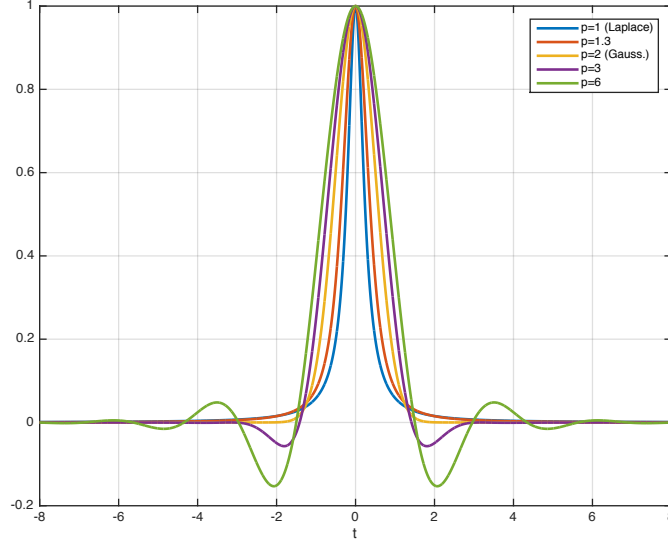


Fig. 1: Plot of the characteristic function of $X \sim \mathcal{N}_p(0, \alpha = 2)$ for several values of p .

A. Analytic Properties of the Characteristic Function

An important question, in particular for numerical methods, is: when the characteristic function of a random variable can be represented as a power series of the form

$$\sum_{k=0}^{\infty} \frac{(it)^k}{k!} \mathbb{E}[X^k]. \quad (19)$$

The above expression is especially useful since the moments of the GG are known for every k , see Proposition 1.

Proposition 4. $\psi_p(t)$ is a real analytic function for

- $t \in \mathbb{R}$ for $p > 1$;
- $|t| < \frac{1}{2}$ for $p = 1$; and
- $t = 0$ for $p < 1$.

Proof: See Appendix D. ■

The above results also lead to the conclusion that for $p > 1$ the moment generating function of $X \sim \mathcal{N}_p(0, 1)$ exists for all t .

B. On Distribution of Zeros of the Characteristic Function

As can be seen from Fig. 1 the characteristic function of the GG distribution can have zeros. The following theorem gives a somewhat surprising result on the distribution of zeros of $\psi_{X,p}(t)$.

Theorem 3. (Distribution of Zeros.) *The characteristic function of $\psi_{X,p}(t)$ has the following properties:*

- for $p > 2$, $\psi_p(t)$ has at least one positive to negative zero crossing; and
- for $0 \leq p \leq 2$, $\psi_p(t)$ is non-negative and can be expressed as

$$\psi_p(t) = \mathbb{E} \left[e^{-\frac{t^2 V^2}{2}} \right], \quad (20)$$

where a random variable V (dependent on p) is defined in Proposition 3.

Proof: See Appendix E. ■

The following result is immediate by Theorem 3.

Note that $\psi_p(\sqrt{2}t)$ can be thought of as a Laplace transform of a random variable V^2 .

Corollary 2. For $0 < p \leq 2$ $\psi_p(t)$ is a strictly decreasing function for $t > 0$.

Proof: The proof follows by taking the derivative of (20)

$$\frac{d}{dt} \psi_p(t) = -t \cdot \mathbb{E} \left[\frac{V^2}{2} e^{-\frac{t^2 V^2}{2}} \right] \quad (21)$$

which is negative for $t > 0$. This concludes the proof. ■

A things which we where not able to show but conjecture to true are

- for $2 < p \leq \infty$ number of zeros is at least countable; and
- for $2 < p < \infty$ zeros do not appear periodically.

C. Deconvolution Results

An important question one can ask is whether the ratio of two characteristic function is still a characteristic function. That is we seek to understand when

$$\psi_{(p,q,\alpha)}(t) = \frac{\psi_q(\alpha \cdot t)}{\psi_p(t)}, \quad (22)$$

is again a characteristic function for all $\alpha \geq 1$. Such a question is important since often we are interested in the existence of random variable X independent of Z such that

$$\alpha Y = X + Z, \quad (23)$$

where Y is the target probability distribution. Here we are interested in answering whether it is possible to transform $Z \sim \mathcal{N}_p(0, 1)$ into αY where $Y \sim \mathcal{N}_q(0, 1)$.

We first focus on the cases when $\psi_{(p,q,\alpha)}(t)$ is not a characteristic function.

Theorem 4. *The function $\psi_{(p,q,\alpha)}(t)$ has the following properties*

- for every $\{(p, q) : q > p > 0\}$ and $\{(p, q) : q \leq 2, p \geq 2, (p, q) \neq (2, 2)\}$, $\psi_{(p,q,\alpha)}(t)$ is not a characteristic function for $\alpha \geq 1$; and
- for every $\{(p, q) : p \geq q > 2\}$ there exists an $\alpha \geq 1$ such that $\psi_{(p,q,\alpha)}(t)$ is not a characteristic function.

Proof: See Appendix F. ■

We would like to point out that for $p \geq q > 2$ there are cases when $\psi_{(p,q,\alpha)}(t)$ is a characteristic function. Specifically, we can find some $\alpha \geq 1$ such that $\psi_{(p,q,\alpha)}(t)$ is a characteristic function $p > 2$ and $q > 2$. The most trivial case is of $p = q$ and $\alpha = 1$ then

$$\psi_{(p,q,\alpha)}(t) = 1, \quad (24)$$

which is a trivial characteristic function. A less trivial example is when $p = q = \infty$ in which case $\psi_{X,\infty}(t) = \text{sinc}(t)$ and

$$\psi_{(\infty,\infty,\alpha)}(t) = \frac{\text{sinc}(\alpha t)}{\text{sinc}(t)}. \quad (25)$$

For example, when $\alpha = 2$ we have that $\psi_{(\infty,\infty,\alpha)}(t) = \frac{1}{2} \cos(2t)$ and corresponds to the characteristic function of random variable ± 1 equally likely. Note that in the above example because zeros of $\psi_p(t)$ occur periodically we can select α such that for $\psi_{(p,q,\alpha)}(t)$ the poles and zeros cancel out.

However, we conjecture that such examples are only possible for $p = \infty$, and for $2 < p < \infty$ zeros of $\psi_p(t)$ do not appear periodically and the second statement of Theorem 4 can be improved to

Conjecture 1. *For $2 < q \leq p < \infty$, $\psi_{(p,q,\alpha)}(t)$ is not a characteristic function for all $\alpha > 1$.*

It is not difficult to check by using analyticity argument used in the proof of Theorem 4 that Conjecture 1 is true if p is even and q is any non-even real number.

One of the consequence of Theorem 4 is that additive transformation from one GG distribution to another GG distribution are not always possible.

Proposition 5. *Let $Z \sim \mathcal{N}_p(0, 1)$*

- *then for $\{(p, q) : q > p > 0\}$ and $\{(p, q) : q \leq 2, p \geq 2, (p, q) \neq (2, 2)\}$ and any $\alpha \geq 1$ there exists no random variable X independent of Z such that*

$$Y = X + Z, \quad (26)$$

where $Y \sim \mathcal{N}_q(0, \alpha^q)$; and

- *then for any $\{(p, q) : p \geq q > 2\}$ there exists some $\alpha \geq 1$ such that there is no random variable X independent of Z such that*

$$Y = X + Z, \quad (27)$$

where $Y \sim \mathcal{N}_q(0, \alpha^q)$.

Proof: The proof of follow trivially from Theorem 4. ■

In view of Theorem 4 it remains to answer what happens to $\psi_{(p,q,\alpha)}(t)$ when $0 < q \leq p < 2$.

Next, we give a sufficient condition for $\psi_{(p,q,\alpha)}(t)$ to be a characteristic function in $\psi_{(p,q,\alpha)}(t)$ when $\{(p, q) : 0 < q \leq p < 2\}$.

Proposition 6. For $\{(p, q) : 0 < q \leq p \leq 2\}$ if $\psi_{(p,q,\alpha)}(t)$ is monotonically decreasing then it is a characteristic function.

Proof: See Appendix G. ■

At this point we are unable to show that $\psi_{(p,q,\alpha)}(t)$ is monotonically decreasing for $\{(p, q) : 0 < q \leq p < 2\}$ even though the numerical simulations seem to suggest so. However, we can solve the following special case.

Proposition 7. For $0 < p = q \leq 2$ the function $\psi_{(p,p,\alpha)}(t)$ is a characteristic function for all $\alpha \geq 1$.

Proof: For the case of $p = q$ by using Theorem 3 we have that there exists some random variable V such that

$$\psi_{(p,p,\alpha)}(t) = \frac{\mathbb{E} \left[e^{-\frac{\alpha^2 t^2 V^2}{2}} \right]}{\mathbb{E} \left[e^{-\frac{t^2 V^2}{2}} \right]}. \quad (28)$$

Clearly, (28) implies that for $\alpha > 1$ the function $\psi_{(p,p,\alpha)}(t)$ is a decreasing function of t and satisfies the condition of Proposition 6. This concludes the proof. ■

For example, in case of $p = 1$ (Laplace) we have that

$$\psi_{(1,1,\alpha)}(t) = \frac{1 + 4t^2}{1 + 4\alpha^2 t^2}, \quad (29)$$

which corresponds to a random variable X with the pdf given by $f_X(x) = \frac{1}{\alpha^2} \delta(x) + (1 - \frac{1}{\alpha^2}) c_1 e^{-\frac{|x|}{2\alpha}}$.

V. INFORMATION MEASURES

A. Entropy and Conditional Entropy

The entropy of the GG distribution is easily evaluate and is given by

Lemma 1. If $X \sim \mathcal{N}_p(0, \alpha^p)$ then

$$h(X) = \frac{1}{p} \ln \left(\frac{pe}{2c_p^p} \mathbb{E}[|X|^p] \right) = \frac{1}{p} \ln \left(\frac{\alpha^p e}{c_p^p} \right). \quad (30)$$

From an information theoretic point of view the class of GG distributions is also interesting since it maximizes the entropy subjects to the absolute p -th moment constraint.

Theorem 5. Let $X \in \mathbb{R}$ such that $\mathbb{E}[|X|^p] \leq \frac{2\alpha^p}{p}$ then

$$h(X) \leq \frac{1}{p} \ln \left(\frac{pe}{2c_p^p} \cdot \mathbb{E}[|X|^p] \right) \leq \frac{1}{p} \ln \left(\frac{\alpha^p e}{c_p^p} \right). \quad (31)$$

The equality is attained iff $X \sim \mathcal{N}_p(0, \alpha^p)$.

Proof. The solution can be given via a method outlined in [13, Chapter 12]. □

The conditional version of Theorem 5 was shown in [15] and is given below.

Theorem 6. For any U such that $h(U) < \infty$ and $\mathbb{E}[|U|^p] < \infty$ for some $p \in (0, \infty)$, and for any V , we have

$$h(U|V) \leq \frac{1}{p} \log \left(\frac{pe}{2c_p^p} \cdot \text{mmpe}(U|V; p) \right), \quad (32)$$

where

$$\text{mmpe}(U|V; p) = \inf_f \mathbb{E}[|U - f(V)|^p], \quad (33)$$

where $f(\cdot)$ is an arbitrary measurable function.

The inequality in Theorem 6 result in a sharper version of the so called continuous version Fano's inequality [13, Chapter 8] that states

$$h(U|V) \leq \frac{1}{2} \log(2\pi e \cdot \text{mmse}(U|V)), \text{ where } \text{mmse}(U|V) = \mathbb{E}[(U - \mathbb{E}[U|V])^2]. \quad (34)$$

Theorem 6 plays a key role in the improvement of the Ozarow-Wyner bound that bound the mutual information of discrete inputs over arbitrary channels [21].

B. Some Regularity Conditions

First we give some technical results on additive noise channels where the noise is distributed according to GG distribution.

Proposition 8. (Regularity of Generalized Gaussian Distribution) For $Z \sim \mathcal{N}_p(0, 1)$ the score function is given by

$$\nabla \log(f_Z(z)) = -\text{sign}(z) \frac{p|z|^{p-1}}{2}, \text{ for } z \neq 0. \quad (35)$$

For the transformation

$$Y = aX + Z, \quad (36)$$

where X and Z are independent we have that

1) In order for $\mathbb{E}[X \cdot \nabla \log(f_Z(Z))|Y]$ to exist it is sufficient that

- $X \in L^1$ for $p = 1$.
- $X \in L^{1+\epsilon}$ for any $\epsilon > 0$ for $p > 1$; and
- $X \in L^{\frac{1}{p}+\epsilon}$ for any $\epsilon > 0$ for $p < 1$.

2) For any X

$$\frac{d}{da} f_Y(y) = \frac{p}{2} f_Y(y) \cdot \mathbb{E} [X \cdot \text{sign}(Z) \cdot |Z|^{p-1} | Y = y], \quad (37)$$

3) For any X

$$\frac{d}{dy} f_Y(y) = \frac{p}{2} f_Y(y) \cdot \mathbb{E} [\text{sign}(Z) \cdot |Z|^{p-1} | Y = y], \quad (38)$$

which implies that the score function of $f_Y(y)$ is given by

$$\frac{d}{dy} \log f_Y(y) = \frac{p}{2} \mathbb{E} [\text{sign}(Z) \cdot |Z|^{p-1} | Y = y]. \quad (39)$$

4) For every a and $X \in L^p$ there exists an integrable $g(y)$ such that

$$|\log(f_Y(y)) \cdot \frac{d}{da} f_Y(y)| \leq g(y). \quad (40)$$

Proof: See Appendix H. ■

C. Mutual Information and Conditional Mean Estimation

The so called I-MMSE relationship [22] has found many application in the contexts of Gaussian noise channels. Next we give a similar expression for channels where the noise is distributed according to GG distribution.

Proposition 9. For any $p \geq 1$ and any $X \in L^p$ independent of Z and $Y = \sqrt{\text{snr}}X + Z$

$$\frac{d}{d\text{snr}} I(X; \sqrt{\text{snr}}X + Z) = M(X, \text{snr}, p), \quad (41)$$

where

$$M(X, \text{snr}, p) = \frac{p}{4\sqrt{\text{snr}}} \mathbb{E} [X \cdot \mathbb{E} [\text{sign}(Z) | Z|^{p-1} | Y]] \quad (42)$$

$$= \frac{p}{4 \cdot \text{snr}} (\mathbb{E} [|Z|^p] - \mathbb{E} [\mathbb{E}[Z|Y] \cdot \mathbb{E} [\text{sign}(Z) | Z|^{p-1} | Y]]) \quad (43)$$

Proof: The result follows by [23] which requires verification of the conditions in Proposition 8. ■

Note that for $p = 2$ by using the orthogonality principle and the identity $(X - \mathbb{E}[X|Y]) = -\frac{1}{\sqrt{\text{snr}}}(Z - \mathbb{E}[Z|Y])$

$$M(X, \text{snr}, 2) = \frac{1}{2} \mathbb{E} [(X - \mathbb{E}[X|Y])^2], \quad (44)$$

which recovers the I-MMSE result.

Lemma 2. $M(X, \text{snr}, p) > 0$ for all $p \geq 1$ and therefore $I(X; Y)$ is an increasing function of snr .

Proof: This follows since $M(X, \text{snr}, p)$ can be re-written as

$$M(X, \text{snr}, p) = \frac{p}{4 \cdot \text{snr}} (\mathbb{E} [|Z|^p] - \mathbb{E} [\mathbb{E}[Z|Y] \cdot \mathbb{E} [\text{sign}(Z) | Z|^{p-1} | Y]]), \quad (45)$$

the non-negativity follows by showing that $\mathbb{E} [\mathbb{E}[Z|Y] \cdot \mathbb{E} [\text{sign}(Z)|Z|^{p-1}|Y]] \leq \mathbb{E} [|Z|^p]$, which is a consequence of repeated use of Jensen's inequality:

$$\mathbb{E} [\mathbb{E}[Z|Y] \cdot \mathbb{E} [\text{sign}(Z)|Z|^{p-1}|Y]] \leq \mathbb{E} [\mathbb{E}[|Z||Y] \cdot \mathbb{E} [|Z|^{p-1}|Y]] \quad (46)$$

$$\leq \mathbb{E} \left[\mathbb{E}^{\frac{1}{p-1}} [|Z|^{p-1}|Y] \cdot \mathbb{E} [|Z|^{p-1}|Y] \right] \quad (47)$$

$$= \mathbb{E} \left[\mathbb{E}^{\frac{p}{p-1}} [|Z|^{p-1}|Y] \right] \quad (48)$$

$$\leq \mathbb{E} [\mathbb{E}[|Z|^p|Y]] \quad (49)$$

$$= \mathbb{E} [|Z|^p]. \quad (50)$$

Note that Jense's inequality holds with equality iff one of the random variables is degenerate. Therefore, all inequalities are strict. This concludes the proof. ■

For $0 < p \leq 2$ we can establish the following relationship between the I-MMSE

Proposition 10. For $0 < p \leq 2$

$$\frac{d}{d\text{snr}} I(X; \sqrt{\text{snr}}X + N|V) = \mathbb{E} \left[\frac{1}{V^2} \text{mmse} \left(X, \frac{\text{snr}}{V^2} \right) \right] \quad (51)$$

where $N = V \cdot Z$.

D. Relative Entropy

Relative Entropy between two GG distributions can be easily computed and is given next.

Proposition 11. For any p and q

$$D(\mathcal{N}_p(0, 1) \| \mathcal{N}_q(0, 1)) = \log \left(\frac{2^{\frac{1}{p} - \frac{1}{q}} c_p}{c_q} \right) + 2^{\frac{q-p}{p}} \frac{\Gamma \left(\frac{q+1}{p} \right)}{\Gamma \left(\frac{1}{p} \right)} - \frac{1}{p}. \quad (52)$$

Another question that often arises in Information Theoretic application is what is the relative entropy distance between $Y = x + Z$ and $Y = X + Z$ where x is a realization of some random variable X .

Proposition 12. Let

$$D(x) = D(\mathcal{N}_p(x, 1) \| f_Y), \quad (53)$$

then for any $p > 0$

$$\frac{D(x_1) - D(x_2)}{x_1 - x_2} = \frac{p}{2} \mathbb{E} [\mathbb{E} [\text{sign}(Z) \cdot |Z|^{p-1} | Y = \bar{Y}]], \quad (54)$$

where $\bar{Y} \sim f_{\bar{Y}}(y) = \frac{F(y-x_2) - F(y-x_1)}{x_1 - x_2}$ where $F(\cdot)$ is the CDF of $\mathcal{N}_p(0, 1)$. Moreover,

$$D'(x) = \frac{p}{2} \mathbb{E} [\mathbb{E} [\text{sign}(Z) \cdot |Z|^{p-1} | Y = \hat{Y}]], \quad (55)$$

where $\hat{Y} \sim \mathcal{N}_p(x, 1)$.

Proof: See Appendix I. ■

Proposition 12 will also be useful later when we demonstrated under which conditions binary communications is optimal in channels with amplitude constraint.

give an example what happens for $p = 2$ and $p = 1$

VI. CHANNEL CAPACITY

A. Shannon's Upper Bound

Using the principle that $X \sim \mathcal{N}_p(0, 1)$ maximizes entropy under p -th moment constraint we arrive at the following upper bound on channel capacity where the input is subject to the p -th moment constraint.

Proposition 13. (Shannon Upper Bound.) For any $p, q > 0$ let

$$C(\alpha; p, q) = \sup_{X \in \mathcal{F}_p(\alpha)} I(X; X + Z), \quad (56a)$$

$$\mathcal{F}_p(\alpha) = \left\{ X : \mathbb{E}[|X|^p] \leq \frac{2\alpha^p}{p} \right\}, \quad (56b)$$

where $Z \sim \mathcal{N}_q(0, 1)$. Then the set $\mathcal{F}_p(\alpha)$ is convex and compact and the supremum in (56a) is achievable for all $p > 0$. Moreover,

$$C(\alpha; p, q) \leq \frac{1}{p} \log(a_{p,q} \cdot \mathbb{E}[|X + Z|^p]), \quad (56c)$$

where $a_{p,q} = \frac{p}{2} \left(\frac{c_q}{c_p}\right)^p e^{\frac{q-p}{q}}$ and the inequality holds iff $Y = X + Z$ is GG distributed.

Proof: See Appendix J. ■

In the above we consider a case of $p = \infty$ to be a case of amplitude constraint that is

$$\mathcal{F}_\infty(\alpha) = \{X : |X| \leq \alpha \text{ a.s.}\}. \quad (57)$$

The upper bound in (56c) is often referred to as Shannon's upper bound. There are several instances when the upper is tight, the most famous of the cases is $(p = 2, q = 2)$, that is Gaussian noise with a second moment constraint. In which case the optimal input distribution is again Gaussian. Next, we look at the case of $(p = 1, q = 1)$ that is Laplace noise and absolute moment constraint.

Proposition 14. For $(p, q) = (1, 1)$ the Shannon's upper bound in (56c) is achievable. Moreover, the pdf of the optimal input distribution is given by $f_X(x) = \frac{1}{4\beta^2} \delta(x) + \left(1 - \frac{1}{4\beta^2}\right) \frac{c_1}{\beta} e^{-\frac{|x|}{2\beta}}$ for some $\beta \in (\alpha, \alpha + 1)$.

Proof: See Appendix K. ■

Comment on the relation to Exponential function [24]

However, in general the Shannon's bound can not be achieved and the inequality in (56c) is actually strict.

Proposition 15.

- The inequality in (56c) is strict for all $\alpha > 0$ and $\{(p, q) : q > p > 0\}$ and $\{(p, q) : q \leq 2, p \geq 2, (p, q) \neq (2, 2)\}$.
- For every $\{(p, q) : p \geq q > 2\}$ there exists and $\alpha > 0$ such that the inequality (56c) is strict.

Proof: Note that the inequality in (56c) is achievable if and only if $Y = X + Z$ is GG. However, by Proposition 5 for the case of $\{(p, q) : q > p > 0\}$ and $\{(p, q) : q \leq 2, p \geq 2, (p, q) \neq (2, 2)\}$ the output Y can not be GG distributed. Similarly, by Proposition 5 for the case of $p \geq q > 2$ there are values of α for which Y can not be distributed according to GG distribution. This concludes the proof. ■

Even-though Shannon's bound is not achievable we can show that it is asymptotically tight. Moreover, we show a remarkable fact that equally space and uniformly distributed input or PAM input achieves a full degree of freedom for any $p, q > 0$.

Proposition 16. For any $p, q > 0$

$$\lim_{\alpha \rightarrow \infty} \frac{C(\alpha; p, q)}{\frac{1}{p} \log(a_{p,q} \cdot \mathbb{E}[|X + Z|^p])} = 1, \quad (58)$$

achieved by PAM with $\mathbb{E}[|X|^p] = \frac{2\alpha^p}{p}$ and number of points given by $N = \left\lfloor \left(\alpha + \frac{p}{2} \mathbb{E}[|Z|^p]\right)^{\frac{1}{p}} \right\rfloor$.

Proof: See Appendix L. ■

B. Achievability with GG input

It is also interesting to point out the discrete inputs are not the only inputs that can get close to capacity, as shown next, this can also be done with inputs from GG family.

Proposition 17. (GG input is not to bad.) For any $p, q > 0$ where $Z \sim \mathcal{N}_q(0, 1)$

$$I(X'; X' + Z) \leq C(\alpha; p, q) \leq I(X'; X' + Z) + C_p, \quad (59)$$

where $X' \sim \mathcal{N}_p(0, \alpha^p)$ and

$$C_p = \begin{cases} \frac{1}{p} \log(2), & 0 \leq p \leq 1, \\ \frac{1}{2} \log(2), & p = 2, \\ \log(2), & \text{else} . \end{cases} \quad (60)$$

Proof: See Appendix M. ■

C. On Capacity Achieving distributions

A possible way of improving on the Shannon's upper bound is to use the duality bound

$$\max_{P_X} I(X; Y) = \max_{P_X} \min_{Q_Y} D(P_{Y|X} \| Q_Y | P_X). \quad (61)$$

However, we in order to do that we must understand the structure of the optimal input distribution.

An interesting questions what type of distribution maximizes the problem in

Proposition 18. (On the Support of the Optimal Input Distribution [25].) For $C(\alpha; p, q)$ we have that

- for $p < q$ capacity achieving distribution has unbounded support; and
- for $p > q$ capacity achieving distribution is compactly supported.

Proof: The proof follows by verifying the condition of [25, Theorem 7]. ■

Another interesting question is when is the input distribution in addition to being compactly supported has a finite (discrete) support.

Proposition 19. For any even integers p and q such that $p > q$ (including the case of $p = \infty$) the maximizing input distribution for $C(\alpha; p, q)$ is discrete with finitely many points.

Proof: The fact that the distribution is compactly support for $p > q$ follows from Proposition 18. The discreteness part follows directly from Smith [26] and requires that the pdf of the noise and the constraint have analytic extensions. However, such analytic extensions are only possible when p and q are even. ■

D. Optimal Input Distribution in a Small Amplitude Regime

Next we investigate a scenario $C(\alpha; \infty, q)$ that is when the input has an amplitude constraint. Specifically, we are interested in identifying when binary communication is optimal. In other words, for what values of α the input $X = \{\pm\alpha\}$ equally likely is optimal. The next result gives necessary and sufficient conditions for this.

Proposition 20. For q even, the input $X = \{\pm\alpha\}$ equally likely is optimal in $C(\alpha; \infty, q)$ if and only if $\alpha \leq \bar{\alpha}_q$ where

$$\bar{\alpha}_q = \sup \left\{ \alpha : \mathbb{E} [\mathbb{E}[Z^{q-1} | Y = \bar{Y}]] \geq 0, \bar{Y} \sim f_{\bar{Y}}(y) = \frac{F(y) - F(y - \alpha)}{\alpha} \right\}, \quad (62)$$

where is $F(\cdot)$ is the CDF of $\mathcal{N}_q(0, 1)$.

Proof: See Appendix N. ■

One can show that for $q = 2$ the condition of Theorem 20 is equivalent to

$$\int \tanh(\bar{\alpha}_2 y) (Q(y - \bar{\alpha}_2) - Q(y)) dy = \frac{\bar{\alpha}_2}{2}, \quad (63)$$

for which the solution is given by $\bar{\alpha}_2 \approx 1.664$. A weaker sufficient condition to that in Theorem 20 can be give by forcing $\mathbb{E}[Z^{p-1} | Y = y]$ to be non-negative for all y which is equivalent to

$$\bar{\alpha}_p = \sup \left\{ \alpha : \frac{(y - \alpha)^{p-1}}{(y + \alpha)^{p-1}} e^{\frac{-(y-\alpha)^p + (y+\alpha)^p}{2}} \leq 1, \forall y > 0 \right\}. \quad (64)$$

APPENDIX A PROOF OF COROLLARY 1

The proof follows by observing that for every $k > 0$ the function

$$g_k(p) = 2^{\frac{k}{p}} \frac{\Gamma\left(\frac{k+1}{p}\right)}{\Gamma\left(\frac{1}{p}\right)} = \mathbb{E}[|X|^k], \quad X \sim \mathcal{N}_p(0, 1). \quad (65)$$

is a decreasing function of p . The second part follow by using Stirling's approximation of the Gamma function $\Gamma(x+1) \approx \sqrt{2\pi x} \left(\frac{x}{e}\right)^x$ and the property that $\Gamma(x+1) = x\Gamma(x)$

$$\begin{aligned}
\left(\frac{2^{\frac{k}{p}-\frac{k}{q}} \Gamma\left(\frac{1}{q}\right) \Gamma\left(\frac{k+1}{p}\right)}{\Gamma\left(\frac{1}{p}\right) \Gamma\left(\frac{k+1}{q}\right)} \right)^{\frac{1}{k}} &= 2^{\frac{1}{p}-\frac{1}{q}} \left(\frac{q\Gamma\left(\frac{1}{q}+1\right) \frac{p}{k+1} \Gamma\left(\frac{k+1}{p}+1\right)}{p\Gamma\left(\frac{1}{p}+1\right) \frac{q}{k+1} \Gamma\left(\frac{k+1}{q}+1\right)} \right)^{\frac{1}{k}} \\
&= 2^{\frac{1}{p}-\frac{1}{q}} \left(\frac{\Gamma\left(\frac{1}{q}+1\right) \Gamma\left(\frac{k+1}{p}+1\right)}{\Gamma\left(\frac{1}{p}+1\right) \Gamma\left(\frac{k+1}{q}+1\right)} \right)^{\frac{1}{k}} \\
&\approx 2^{\frac{1}{p}-\frac{1}{q}} \left(\frac{\left(\frac{1}{qe}\right)^{\frac{1}{q}} \cdot \left(\frac{k+1}{pe}\right)^{\frac{k+1}{p}}}{\left(\frac{1}{pe}\right)^{\frac{1}{p}} \cdot \left(\frac{k+1}{qe}\right)^{\frac{k+1}{q}}} \right)^{\frac{1}{k}} \\
&= 2^{\frac{1}{p}-\frac{1}{q}} \left(e^{\frac{1}{p}-\frac{1}{q}} e^{\frac{k+1}{q}-\frac{k+1}{p}} \frac{q^{\frac{k}{q}}}{p^{\frac{k}{p}}} \cdot (k+1)^{\frac{k+1}{p}-\frac{k+1}{q}} \right)^{\frac{1}{k}} \\
&= 2^{\frac{1}{p}-\frac{1}{q}} e^{\frac{1}{q}-\frac{1}{p}} \frac{q^{\frac{1}{q}}}{p^{\frac{1}{p}}} (k+1)^{\frac{k+1}{k}(\frac{1}{p}-\frac{1}{q})}.
\end{aligned}$$

The proof is concluded by taking the limit at $k \rightarrow \infty$ and using $q > p$.

APPENDIX B PROOF OF PROPOSITION 1

The proof that $e^{-\frac{|x|^p}{2}}$ is positive definite for $0 < p \leq 2$ was addressed in [27] and [1] where also the expression in (13) was shown.

To show that $e^{-\frac{|x|^p}{2}}$ is not positive definite for $p > 2$ it is enough to consider the following counter example. In the Definition 1 let $n = 3$ and choose $|x_1 - x_2| = \epsilon, |x_2 - x_3| = a\epsilon, |x_1 - x_3| = (a+1)\epsilon$ for some $\epsilon, a > 0$. Therefore, the determinant of A is given by

$$\begin{aligned}
h(\epsilon) &:= |A| = 1 - e^{-\frac{2a^p\epsilon^p}{2}} - e^{-\frac{\epsilon^p}{2}} \left(e^{-\frac{\epsilon^p}{2}} - e^{-\frac{(a^p+(a+1)^p)\epsilon^p}{2}} \right) + e^{-\frac{(a+1)^p\epsilon^p}{2}} \left(e^{-\frac{(a^p+1)\epsilon^p}{2}} - e^{-\frac{(a+1)^p\epsilon^p}{2}} \right) \\
&= 1 - e^{-\frac{2a^p\epsilon^p}{2}} - e^{-\frac{2\epsilon^p}{2}} + 2e^{-\frac{((a+1)^p+a^p+1)\epsilon^p}{2}} - e^{-\frac{2(a+1)^p\epsilon^p}{2}}.
\end{aligned}$$

The idea is to show that for a small ϵ we have that $h(\epsilon) < 0$. To that end, we use the following approximation $e^t = 1 + t + \frac{t^2}{2} + O(t^3)$ and

$$\begin{aligned}
h(\epsilon) &= 1 - \left(1 - \frac{2a^p\epsilon^p}{2} + \left(\frac{2a^p\epsilon^p}{2} \right)^2 \right) - \left(1 - \frac{2\epsilon^p}{2} + \left(\frac{2\epsilon^p}{2} \right)^2 \right) - \left(1 - \frac{2(a+1)^p\epsilon^p}{2} + \left(\frac{2(a+1)^p\epsilon^p}{2} \right)^2 \right) \\
&\quad + 2 \left(1 - \frac{((a+1)^p+a^p+1)\epsilon^p}{2} + \left(\frac{((a+1)^p+a^p+1)\epsilon^p}{2} \right)^2 \right) + O(\epsilon^{3p})
\end{aligned} \tag{66}$$

$$= \epsilon^{2p} \left(\frac{((a+1)^p+a^p+1)^2}{2} - a^{2p} - (a+1)^{2p} - 1 \right) + O(\epsilon^{3p}). \tag{67}$$

The proof is concluded by taking ϵ small enough and noting that $\frac{((a+1)^p+a^p+1)^2}{2} - a^{2p} - (a+1)^{2p} - 1 \geq 0$ for $p \leq 2$ and $\frac{((a+1)^p+a^p+1)^2}{2} - a^{2p} - (a+1)^{2p} - 1 < 0$ for $p > 2$.

APPENDIX C
PROOF OF PROPOSITION 3

To show that $X = V \cdot Z$, first observe that, $d\nu(t) = c_p \sqrt{\frac{2\pi}{t}} d\mu(t)$ is a probability measure and where $d\mu(t)$ was defined in Theorem 1

$$1 = \mathbb{P}(X \in \mathbb{R}) = \int_{\mathbb{R}} c_p e^{-\frac{|x|^p}{2}} dx \quad (68)$$

$$= \int_{\mathbb{R}} c_p \int_0^\infty e^{-\frac{tx^2}{2}} d\mu(t) dx \quad (69)$$

$$= c_p \int_0^\infty \int_{\mathbb{R}} e^{-\frac{tx^2}{2}} dx d\mu(t) \quad (70)$$

$$= c_p \int_0^\infty \sqrt{\frac{2\pi}{t}} d\mu(t) \quad (71)$$

$$= \int_0^\infty d\nu(t). \quad (72)$$

The above implies that $d\nu(t) = c_p \sqrt{\frac{2\pi}{t}} d\mu(t)$ is a probability measure on $[0, \infty)$.

Moreover, for any measurable set $\mathcal{S} \subset \mathbb{R}$ we have

$$\mathbb{P}(X \in \mathcal{S}) = \int_{\mathcal{S}} c_p e^{-\frac{|x|^p}{2}} dx \quad (73)$$

$$\stackrel{a)}{=} \int_{\mathcal{S}} c_p \int_0^\infty e^{-\frac{tx^2}{2}} d\mu(t) dx \quad (74)$$

$$\stackrel{b)}{=} \int_0^\infty \int_{\mathcal{S}} e^{-\frac{tx^2}{2}} dx c_p d\mu(t) \quad (75)$$

$$= \int_0^\infty \int_{\mathcal{S}} \sqrt{\frac{t}{2\pi}} e^{-\frac{tx^2}{2}} dx c_p \sqrt{\frac{2\pi}{t}} d\mu(t) \quad (76)$$

$$= \int_0^\infty \mathbb{P}\left(\frac{1}{\sqrt{t}} Z \in \mathcal{S}\right) c_p \sqrt{\frac{2\pi}{t}} d\mu(t) \quad (77)$$

$$\stackrel{c)}{=} \mathbb{E}\left[\mathbb{P}\left(\frac{1}{\sqrt{t}} Z \in \mathcal{S}\right)\right] \quad (78)$$

$$= \mathbb{P}\left(\frac{1}{\sqrt{T}} \cdot Z \in \mathcal{S}\right) \quad (79)$$

$$\stackrel{d)}{=} \mathbb{P}(V \cdot Z \in \mathcal{S}), \quad (80)$$

where equalities follow from: a) using the representation of $e^{-\frac{|x|^p}{2}}$ in Theorem 1; b) Tonelli's theorem for non-negative functions; c) using that $c_p \sqrt{\frac{2\pi}{t}} d\mu(t)$ is a probability measure; and d) renaming $V = \frac{1}{\sqrt{T}}$.

Therefore, it follows that X can be decomposed into $V \cdot Z$.

Next, we show that for $p < 2$ V is an unbounded random variable.

Any random variable V is unbounded iff

$$\lim_{k \rightarrow \infty} \mathbb{E}^{\frac{1}{k}}[V^k] = \infty. \quad (81)$$

To show that V is unbounded observe that all the moments of V are given by

$$\mathbb{E}[V^k] = \frac{\mathbb{E}[|X|^k]}{\mathbb{E}[|Z|^k]}, \quad k \in \mathbb{R}^+, \quad (82)$$

where $X \sim \mathcal{N}_p(0, 1)$ and $Z \sim \mathcal{N}_2(0, 1)$. Since, we are assuming that $p < 2$ by Corollary 1 we have that

$$\lim_{k \rightarrow \infty} \mathbb{E}^{\frac{1}{k}}[V^k] = \lim_{k \rightarrow \infty} \left(\frac{\mathbb{E}[|X|^k]}{\mathbb{E}[|Z|^k]} \right)^{\frac{1}{k}} = \infty. \quad (83)$$

Therefore, V is an unbounded random variable.

Next, we check when the characteristic function of V can be expressed as a power series and compute the radius of convergence r given by

$$\frac{1}{r} = \lim_{k \rightarrow \infty} \left(\frac{\mathbb{E}[V^k]}{k!} \right)^{\frac{1}{k}}. \quad (84)$$

Using the steps leading to the proof of Corollary 1 we have that

$$\mathbb{E}[V^k] \approx \left(\frac{2}{e} \right)^{\frac{k}{p} - \frac{k}{2}} \frac{2^{\frac{k}{2}}}{p^{\frac{k}{p}}} (k+1)^{(k+1)(\frac{1}{p} - \frac{1}{2})}. \quad (85)$$

Using (85) we have that the radius of convergence r is given by

$$\frac{1}{r} = \lim_{k \rightarrow \infty} \left(\frac{\mathbb{E}[V^k]}{k!} \right)^{\frac{1}{k}} \quad (86)$$

$$= \left(\frac{2}{e} \right)^{\frac{1}{p} - \frac{1}{2}} \frac{2^{\frac{1}{2}}}{p^{\frac{1}{p}}} \lim_{k \rightarrow \infty} \frac{(k+1)^{\frac{(k+1)}{k}(\frac{1}{p} - \frac{1}{2})}}{k^{\frac{k+1}{k}}}. \quad (87)$$

$$= \begin{cases} \infty, & p < \frac{2}{3} \\ 0, & p > \frac{2}{3} \end{cases}. \quad (88)$$

This concludes the proof.

APPENDIX D PROOF OF PROPOSITION 4.

To show that $\psi_p(t)$ can be represented by the power series we perform a ratio test and compute the radius of convergence, as follows

$$r = \lim_{k \rightarrow \infty} \frac{\frac{\mathbb{E}[|X|^k]}{k!}}{\frac{\mathbb{E}[|X|^{k+1}]}{(k+1)!}} \quad (89)$$

$$= \lim_{k \rightarrow \infty} \frac{k 2^{-\frac{1}{p}} \Gamma\left(\frac{k+1}{p}\right)}{\Gamma\left(\frac{k+2}{p}\right)} \quad (90)$$

$$= 2^{-\frac{1}{p}} \lim_{k \rightarrow \infty} \frac{k \frac{p}{k+1} \Gamma\left(\frac{k+1}{p} + 1\right)}{\frac{p}{k+2} \Gamma\left(\frac{k+2}{p} + 1\right)} \quad (91)$$

$$= 2^{-\frac{1}{p}} \lim_{k \rightarrow \infty} \frac{k \Gamma\left(\frac{k+1}{p} + 1\right)}{\Gamma\left(\frac{k+2}{p} + 1\right)}. \quad (92)$$

Now for $p = 1$ the limit in (92) can be computed as follows

$$\lim_{k \rightarrow \infty} \frac{k \Gamma(k+2)}{\Gamma(k+3)} = \lim_{k \rightarrow \infty} \frac{k \Gamma(k+2)}{(k+2) \Gamma(k+2)} = 1. \quad (93)$$

Therefore, for $p = 1$ we have that $r = 1/2$.

For $p \neq 1$ the limit in (92) can be compute using Stirling's approximation as follows

$$\lim_{k \rightarrow \infty} \frac{k \Gamma\left(\frac{k+1}{p} + 1\right)}{\Gamma\left(\frac{k+2}{p} + 1\right)} = \lim_{k \rightarrow \infty} \frac{k \sqrt{k+1} \left(\frac{k+1}{ep}\right)^{\frac{k+1}{p}}}{\sqrt{k+2} \left(\frac{k+2}{ep}\right)^{\frac{k+2}{p}}} \quad (94)$$

$$= (ep)^{\frac{1}{p}} \lim_{k \rightarrow \infty} \frac{k (k+1)^{\frac{k+1}{p}}}{(k+2)^{\frac{k+2}{p}}} \quad (95)$$

$$= (ep)^{\frac{1}{p}} \lim_{k \rightarrow \infty} k^{\frac{p-1}{p}} = \begin{cases} \infty & p > 1 \\ 0 & p < 1 \end{cases}. \quad (96)$$

This concludes the proof.

APPENDIX E
PROOF OF THEOREM 3

We first show that for $p > 2$ there is at least one zero. We use the approach of [28]. Towards a contradiction assume that $\psi_p(t) \geq 0$ for all $t \geq 0$, then for all $t \geq 0$ we have that

$$0 \leq \frac{1}{2c_p} \frac{1}{2\pi} \int_0^\infty \psi_p(x) (1 - \cos(xt))^2 dx \quad (97)$$

$$\stackrel{a)}{=} \frac{1}{2c_p} \frac{1}{2\pi} \int_0^\infty \psi_p(x) \frac{1}{2} (3 - 4\cos(tx) + \cos(2tx)) dx \quad (98)$$

$$\stackrel{b)}{=} 3 - 4e^{-\frac{t^p}{2}} + e^{-\frac{(2t)^p}{2}}, \quad (99)$$

where equalities follow from: a) using $(1 - \cos(xt))^2 = \frac{1}{2} (3 - 4\cos(tx) + \cos(2tx))$; and b) using inverse Fourier transform and Parseval's identity. For small x we can write $e^{-x} = 1 - x + O(x^2)$ and therefore we have that

$$0 \leq 3 - 4 \left(1 - \frac{t^p}{2}\right) + \left(1 - \frac{(2t)^p}{2}\right) + O(t^{2p}) = (4 - 2^p) \frac{t^p}{2} + O(t^{2p}), \quad (100)$$

Therefore, for $p > 2$ we reach a contradiction since $4 - 2^p < 0$ for $p > 2$. This concludes the proof for the case of $p > 2$.

For $0 < p \leq 2$ goes as follows

$$\psi_p(t) = \mathbb{E}[e^{itX}] \quad (101)$$

$$\stackrel{a)}{=} \mathbb{E}[e^{itVZ}] \quad (102)$$

$$= \mathbb{E}[\mathbb{E}[e^{itVZ} | V]] \quad (103)$$

$$\stackrel{b)}{=} \mathbb{E}\left[e^{-\frac{t^2 V^2}{2}}\right] \quad (104)$$

$$\stackrel{c)}{>} 0, \quad (105)$$

where (in)-equalities follow from: a) the decomposition in Proposition ??; b) the independence of V and Z and the fact that characteristic function of Z is $e^{-\frac{t^2}{2}}$; and c) positivity of $e^{-\frac{t^2 V^2}{2}}$.

This concludes the proof.

APPENDIX F
PROOF OF THEOREM 4

We start by looking at the regime $q > p > 0$. We want to show that there exists no random variable X independent of $Z \sim \mathcal{N}_p(0, 1)$ such that

$$\alpha Y = X + Z, \quad (106)$$

where $Y \sim \mathcal{N}_q(0, 1)$ for all $\alpha \geq 1$. Note that since Y and Z are symmetric and have finite moments if such an X exists it must also be symmetric with finite moments.

Then for all $k \geq 1$

$$\alpha^k \mathbb{E}[|Y|^k] = \mathbb{E}[|X + Z|^k] \quad (107)$$

$$= \mathbb{E}[\mathbb{E}[|X + Z|^k | Z]] \quad (108)$$

$$\stackrel{a)}{\geq} \mathbb{E}[\mathbb{E}[|X + Z|^k | Z]] \quad (109)$$

$$\stackrel{b)}{=} \mathbb{E}[\mathbb{E}[X] + |Z|^k] \quad (110)$$

$$= \mathbb{E}[|Z|^k], \quad (111)$$

where the (in)-equalities follow from: a) Jensen's inequality; and b) independence of X and Z .

This in turn implies that in order for inequality in (111) to be true we must have that

$$\alpha \geq \left(\frac{\mathbb{E}[|Z|^k]}{\mathbb{E}[|Y|^k]} \right)^{\frac{1}{k}}, \quad (112)$$

for all $k \geq 1$. However, by Corollary 1 for $p < q$

$$\lim_{k \rightarrow \infty} \left(\frac{\mathbb{E}[|Z|^k]}{\mathbb{E}[|Y|^k]} \right)^{\frac{1}{k}} = \infty, \quad (113)$$

therefore, there exists no $\alpha \geq 1$ that can satisfy (112). This concludes the proof for $p < q$.

Next we focus on the regime $p \geq 2 \geq q$ and $p \neq q$.

Note that in the case of $p = 2$ and $q < 2$ we want to show that there is no random variable X such that the convolution leads to

$$f_Y(y) = c_2 \mathbb{E} \left[e^{-\frac{(y-X)^2}{2}} \right], \text{ where } f_Y(y) = \frac{c_q}{\alpha} e^{-\frac{|y|^q}{2\alpha^q}}. \quad (114)$$

Such X does not exist since the convolution preserves analyticity. That is the convolution of with an analytic pdf must result in an analytic pdf. Noting that $f_Y(y)$ is not analytic for $q < 2$ leads to the desired conclusion.

Now for $p > 2$ and $q \leq 2$ the function $\psi_{(p,q,\alpha)}(t)$ has a pole but no zeros by Theorem 3. Therefore, for the case of $p > 2$ and $q \leq 2$ there exists a t_0 , namely the pole of $\psi_{(p,q,\alpha)}(t)$, such that $\psi_{(p,q,\alpha)}(t)$ is not continuous at $t = t_0$. This violates the condition that the characteristic function is always a continuous function of t and therefore $\psi_{(p,q,\alpha)}(t)$ is not a characteristic function for all $\alpha \geq 1$.

Next we focus on the case of $p > 2$ and $q > 2$. However, since the case of $p < q$ was already handled it is enough to look at the case of $p \geq q > 2$. For the case of $p > 2$ and $q > 2$ the function

$$\psi_{(p,q,\alpha)}(t) = \frac{\psi_p(\alpha t)}{\psi_q(t)}, \quad (115)$$

has both poles and zeros by Theorem 3. Moreover, t_1 be such that is $\psi_q(t_1) = 0$. Therefore, we can always choose an α such that $\psi_p(\alpha t_1) \neq 0$ and

$$\psi_{(p,q,\alpha)}(t_1) = \infty. \quad (116)$$

In other words, we choose an α such that the poles do not cancel the zeros. This implies that there exists an α such that $\psi_{(p,q,\alpha)}(t)$ is not a continuous function of t and is not a characteristic function.

This concludes the proof.

APPENDIX G PROOF OF PROPOSITION 6

For our argument we will need the *second mean value theorem* for integration which states that for a monotonically decreasing $f : [a, b] \rightarrow \mathbb{R}$ and integrable $g : [a, b] \rightarrow \mathbb{R}$ we have that there exists some $c \in (a, b)$ such that

$$\int_a^b f(x)g(x)dx = f(a) \int_a^c g(x)dx. \quad (117)$$

By Cramer's theorem [29] $\psi_{(p,q,\alpha)}(t)$ is a characteristic function if and only if it is bounded, continuous and satisfies

$$\int_0^A \int_0^A \psi_{(p,q,\alpha)}(t-s) e^{ix(t-s)} dt ds \geq 0. \quad (118)$$

for all $A \geq 0$ and all $x \in \mathbb{R}$. By symmetry of $\psi_{(p,q,\alpha)}(t)$ the criteria in the equation (118) can be simplified to

$$\int_0^A \int_0^A \psi_{(p,q,\alpha)}(t-s) e^{ix(t-s)} dt ds \stackrel{a)}{=} 2 \int_0^A \int_0^s \psi_{(p,q,\alpha)}(t-s) \cos(x(t-s)) dt ds \quad (119)$$

$$\stackrel{b)}{=} 2 \int_0^A (A-u) \psi_{(p,q,\alpha)}(u) \cos(x \cdot u) du. \quad (120)$$

where the equalities follow from: a) symmetry around $t = s$; and b) the change of variable.

Note that for the case of $x = 0$ the (118) is satisfied. Moreover, since $\cos(\cdot)$ is an even function we can restrict our attention to $x > 0$ and by another change of variable we have that (118) is equivalent to

$$\frac{2}{x} \int_0^{xA} \left(A - \frac{u}{x}\right) \psi_{(p,q,\alpha)}\left(\frac{u}{x}\right) \cos(u) du = \frac{2}{x^2} \int_0^{xA} (xA - u) \psi_{(p,q,\alpha)}\left(\frac{u}{x}\right) \cos(u) du \geq 0, \quad (121)$$

for $x > 0$ and $A \geq 0$.

Next by using the assumption that $\psi_{(p,q,\alpha)}\left(\frac{u}{x}\right)$ is a monotonically decreasing then for $x A \leq \pi$ by *second mean value theorem for integration* we have that

$$\int_0^{xA} (xA - u) \psi_{(p,q,\alpha)}\left(\frac{u}{x}\right) \cos(u) du = \psi_{(p,q,\alpha)}(0) \int_0^b \left(A - \frac{u}{x}\right) \cos(u) du, \quad (122)$$

for some $b \in [0, xA]$. Note that the integral $\int_0^b (xA - u) \cos(u) du \geq 0$ for all $b \in [0, xA]$. For $xA > \pi$ let K be the largest integer such that $xA \geq K\pi$ then

$$\int_0^{\frac{A}{x}} (xA - u) \psi_{(p,q,\alpha)}\left(\frac{u}{x}\right) \cos(u) du \quad (123)$$

$$= \sum_{k=0}^{K-1} \int_{k\pi}^{(k+1)\pi} (xA - u) \psi_{(p,q,\alpha)}\left(\frac{u}{x}\right) \cos(u) du + \int_{K\pi}^{xA} (xA - u) \psi_{(p,q,\alpha)}\left(\frac{u}{x}\right) \cos(u) du \quad (124)$$

$$= \sum_{k=0}^{K-1} \int_0^\pi (xA - v - k\pi) \psi_{(p,q,\alpha)}\left(\frac{v + k\pi}{x}\right) \cos(v + k\pi) dv + \int_0^{xA-K\pi} (xA - v - K\pi) \psi_{(p,q,\alpha)}\left(\frac{v + K\pi}{x}\right) \cos(v + K\pi) dv \quad (125)$$

$$\stackrel{a)}{=} \sum_{k=0}^{K-1} \psi_{(p,q,\alpha)}\left(\frac{k\pi}{x}\right) \int_0^\pi (xA - v - k\pi) \cos(v + k\pi) dv + \psi_{(p,q,\alpha)}\left(\frac{K\pi}{x}\right) \int_0^{xA-K\pi} (xA - v - K\pi) \cos(v + K\pi) dv \quad (126)$$

$$= \sum_{k=0}^{K-1} \psi_{(p,q,\alpha)}\left(\frac{k\pi}{x}\right) (-1)^k \int_0^\pi (xA - v - k\pi) \cos(v) dv + (-1)^K \psi_{(p,q,\alpha)}\left(\frac{K\pi}{x}\right) \int_0^{xA-K\pi} (xA - v - K\pi) \cos(v) dv \quad (127)$$

$$\stackrel{b)}{=} 2 \sum_{k=0}^{K-1} (-1)^k \psi_{(p,q,\alpha)}\left(\frac{k\pi}{x}\right) + (-1)^K \psi_{(p,q,\alpha)}\left(\frac{K\pi}{x}\right) (1 - \cos(xA - K\pi)) \quad (128)$$

where equalities follow from: a) second mean value theorem for integration; and b) observing that $\int_0^\pi (xA - v - k\pi) \cos(v) dv = 2$ and $\int_0^{xA-K\pi} (xA - v - K\pi) \cos(v) dv = 1 - \cos(xA - K\pi)$.

Therefore, it remains to show that (128) is non-negative.

Note that the first sum of (128) is positive for all K and for even K the sum in (128) is positive.

For K odd using that $\psi_{(p,q,\alpha)}(y)$ is monotonically decreasing we have that the first sum of (128) can be lower bounded as follows

$$\sum_{k=0}^{K-1} (-1)^k \psi_{(p,q,\alpha)}\left(\frac{k\pi}{x}\right) \geq \psi_{(p,q,\alpha)}\left(\frac{(K-1)\pi}{x}\right) \sum_{k=0}^{K-1} (-1)^k = \psi_{(p,q,\alpha)}\left(\frac{(K-1)\pi}{x}\right), \quad (129)$$

and we have that (128) is lower bounded by

$$2\psi_{X,(p,q)}\left(\frac{(K-1)\pi}{x}\right) - \psi_{(p,q,\alpha)}\left(\frac{K\pi}{x}\right) (1 - \cos(xA - K\pi)) \geq \psi_{(p,q,\alpha)}\left(\frac{K\pi}{x}\right) (1 + \cos(xA - K\pi)) \geq 0 \quad (130)$$

which concludes the proof.

APPENDIX H PROOF OF PROPOSITION 8

A sufficient conditions for the existence of conditional expectation is existence of unconditional expectation. That is to show that $\mathbb{E}[X \cdot \nabla \log(f_Z(Z))|Y]$ exists we have to show that $\mathbb{E}[X \cdot \nabla \log(f_Z(Z))]$ is finite.

For the case of $p = 1$ observe that

$$\mathbb{E}[X \cdot \nabla \log(f_Z(Z))] \leq \mathbb{E}[|X \cdot \nabla \log(f_Z(Z))|] \quad (131)$$

$$= \frac{p}{2} \mathbb{E}[|X \cdot \text{sign}(Z)|] \quad (132)$$

$$= \frac{p}{2} \mathbb{E}[|X|] \quad (133)$$

thus for the existence of $\mathbb{E}[X \cdot \nabla \log(f_Z(Z))|Y]$ it is sufficient that $\mathbb{E}[|X|] < \infty$.

For the case of $p \neq 1$ observe that

$$\mathbb{E}[X \cdot \nabla \log(f_Z(Z))] \leq \mathbb{E}[|X \cdot \nabla \log(f_Z(Z))|] \quad (134)$$

$$\leq \frac{p}{2} \mathbb{E}[|X \cdot \text{sign}(Z)| |Z|^{p-1}] \quad (135)$$

$$= \frac{p}{2} \mathbb{E}[|X| \cdot |Z|^{p-1}] \quad (136)$$

$$\leq \frac{p}{2} \mathbb{E}^{\frac{1}{m}}[|X|^m] \mathbb{E}^{\frac{1}{r}}[|Z|^{(p-1)r}] \quad (137)$$

where the last inequality is due to Holder's inequality. By Proposition 1 for any $p > 0$, $\mathbb{E}^{\frac{1}{r}} [|Z|^{(p-1)r}]$ is finite if $-1 < (p-1)r < \infty$. Moreover, by using $1 = \frac{1}{m} + \frac{1}{r}$ we have that $\mathbb{E}^{\frac{1}{r}} [|Z|^{(p-1)r}]$ is finite if

$$\frac{1}{m} < p < \infty, \quad (138)$$

Therefore, for $p < 1$ in order for (137) to be finite we require that $\mathbb{E}[|X|^m] < \infty$ for $m > \frac{1}{p}$. For $p > 1$ in order for (137) we require that $\mathbb{E}[|X|^m] < \infty$ for any $m > \frac{1}{p}$, since $p > 1$ we can take $m > 1$.

To show (37) we observe that

$$\frac{d}{da} f_Y(y) = \frac{d}{da} \mathbb{E}[f_Z(y - aX)] = \mathbb{E} \left[\frac{d}{da} f_Z(y - aX) \right] = \frac{p}{2} \cdot \mathbb{E} [X \cdot \text{sign}(y - aX) \cdot |y - aX|^{p-1} \cdot f_Z(y - aX)], \quad (139)$$

To justify differentiation under integration we have to show that there exists integrable function $\theta(X)$ such that $|\frac{d}{da} f_Z(y - aX)| \leq \theta(x)$ for all a .

$$\left| \frac{d}{da} f_Z(y - aX) \right| = |X \cdot \text{sign}(y - aX) \cdot |y - aX|^{p-1} \cdot f_Z(y - aX)| \quad (140)$$

$$= |X| \cdot |y - aX|^{p-1} \cdot f_Z(y - aX) \quad (141)$$

$$\leq |X| \cdot 2^{p-1} (|y|^{p-1} + |aX|^{p-1}) \cdot f_Z(y - aX) \quad (142)$$

$$= |y|^{p-1} 2^{p-1} |X| \cdot f_Z(y - aX) + |a|^{p-1} 2^{p-1} |X|^p f_Z(y - aX) = \theta(X), \quad (143)$$

note that because $f_Z(y - aX) \sim e^{-\frac{|X|^p}{2}}$ we have that $\theta(X)$ is integrable for any y, p and a . The proof for $\frac{d}{dy} f_Y(y)$ follows similarly.

To verify (40) observe that by Jensen's inequality

$$f_Y(y) = \mathbb{E} \left[c_p e^{-\frac{|y-aX|^p}{2}} \right] \quad (144)$$

$$\geq c_p e^{-\mathbb{E} \left[\frac{|y-aX|^p}{2} \right]} \quad (145)$$

$$\geq c_p e^{-\frac{2^p (|y|^p + |a|^p \mathbb{E}[|X|^p])}{2}}, \quad (146)$$

and since $f_Y(y) = \mathbb{E} \left[c_p e^{-\frac{|y-aX|^p}{2}} \right] \leq \mathbb{E} [c_p e^0] \leq c_p \leq 1$ we have that $\log(f_Y(y)) \leq 0$ and

$$|\log(f_Y(y))| \leq \frac{2^p (|y|^p + |a|^p \mathbb{E}[|X|^p])}{2} + |\log(c_p)|. \quad (147)$$

Therefore,

$$|\log(f_Y(y)) \cdot \frac{d}{da} f_Y(y)| \leq \left(\frac{2^p (|y|^p + |a|^p \mathbb{E}[|X|^p])}{2} + |\log(c_p)| \right) \theta(X), \quad (148)$$

where $\theta(X)$ is defined in (143). Thus, for (40) to hold we require that $\mathbb{E}[|X|^p] < \infty$. This concludes the proof.

APPENDIX I PROOF OF PROPOSITION 12

$$D(x_1) - D(x_2) = - \int \log(f_Y(y)) \cdot \left(c_p e^{-\frac{|y-x_1|^p}{2}} - c_p e^{-\frac{|y-x_2|^p}{2}} \right) dy \quad (149)$$

$$= \int \nabla \log(f_Y(y)) \cdot (F(y - x_2) - F(y - x_1)) dy \quad (150)$$

where the last equation follows from integration by parts. Note that

$$\int F(y - x_2) - F(y - x_1) dy = x_1 - x_2. \quad (151)$$

Therefore,

$$D(x_1) - D(x_2) = (x_1 - x_2) \mathbb{E} [\nabla \log(f_Y(\bar{Y}))], \quad (152)$$

where $\bar{Y} \sim f_Y(y) = \frac{F(y-x_2) - F(y-x_1)}{x_1 - x_2}$. Moreover, using (38) we have that

$$D(x_1) - D(x_2) = \frac{p}{2} (x_1 - x_2) \mathbb{E} [\mathbb{E} [\text{sign}(Z) \cdot |Z|^{p-1} |Y = \bar{Y}]]]. \quad (153)$$

This also implies that

$$\frac{d}{dx} D(x) = \frac{p}{2} \mathbb{E} [\mathbb{E} [\text{sign}(Z) \cdot |Z|^{p-1} |Y = \hat{Y}]]], \quad (154)$$

where $\hat{Y} \sim \mathcal{N}_p(x, 1)$. This concludes the proof.

APPENDIX J
PROOF OF PROPOSITION 13

The convexity of $\mathcal{F}_p(\alpha)$ follows trivially. To show that $\mathcal{F}_p(\alpha)$ is compact it is enough to show that it is closed and tight. To show that the set $\mathcal{F}_p(\alpha)$ is tight we need to show that for any ϵ there exists a_ϵ such that

$$\mathbb{P}[|X| \geq a_\epsilon] \leq \epsilon. \quad (155)$$

This follows by Chebyshev's inequality, since for every $p > 0$

$$\mathbb{P}[|X| \geq a_\epsilon] \leq \frac{2\alpha^p}{pa_\epsilon^p}, \quad (156)$$

therefore selecting a_ϵ such that $\frac{2\alpha^p}{pa_\epsilon^p} = \epsilon$ show that the set is tight. The fact that the set is closed follows by weak convergence since

$$\int f dF \leq \liminf_{n \rightarrow \infty} \int f dF_n, \quad (157)$$

for every function f bounded from below.

The proof of the upper bound follows using a classical approach

$$I(X; X + Z) = h(X + Z) - h(Z) \quad (158)$$

$$\leq \frac{1}{p} \ln \left(\frac{pe}{2c_p^p} \cdot \mathbb{E}[|X + Z|^p] \right) - \frac{1}{q} \ln \left(\frac{qe}{2c_q^q} \cdot \mathbb{E}[|Z|^q] \right), \quad (159)$$

where the inequality follows by using the fact that GG distribution maximize the entropy under p -th moment constraint.

This concludes the proof.

APPENDIX K
PROOF OF PROPOSITION 14

Note that the Shannon upper bound is achievable if and only if $Y \sim \mathcal{N}_q(0, \beta)$ where according to Proposition 1 $\beta = \frac{1}{2} \mathbb{E}[|X + Z|]$ for some $X \in \mathcal{F}_p$ where

$$\beta = \frac{1}{2} \mathbb{E}[|X + Z|] \leq \frac{1}{2} (2\alpha + 2) = \alpha + 1. \quad (160)$$

Next, we must demonstrated that there exist X such that $Y \sim \mathcal{N}_1(0, \beta)$ and $X \in \mathcal{F}_1(\alpha)$. Using Proposition 6 we know that there exists a input distribution X such that $Y \sim \mathcal{N}_q(0, \beta)$ with the characteristic function given by

$$\phi_X(t) = \frac{1 + 4t^2}{1 + 4\beta^2 t^2}. \quad (161)$$

Therefore, it remains to show that $X \in \mathcal{F}_1(\alpha)$. In other words, we wan to show that $\mathbb{E}[|X|] \leq 2\alpha$. Since, we know the form of the characteristic function of X we can use the result of [30] to compute the first absolute moment of X as follows

$$\begin{aligned} \mathbb{E}[|X|] &\stackrel{a)}{=} -\frac{1}{\pi} \int_0^\infty \frac{\phi^{(1)}(t)}{t} dt \\ &= -\frac{1}{\pi} \int_0^\infty \left(\frac{-8(\beta^2 - 1)t}{(1 + 4\beta^2 t^2)^2} \right) \frac{1}{t} dt \\ &= \frac{\beta^2 - 1}{\beta} \\ &\stackrel{b)}{\leq} \frac{(\alpha + 1)^2 - 1}{(\alpha + 1)} \\ &= \alpha \frac{\alpha + 2}{\alpha + 1} \\ &\leq 2\alpha, \end{aligned}$$

where (in)-equalities follow from: a) using [30, Theorem 1]; and b) using the fact that $\frac{\beta^2 - 1}{\beta}$ is increasing function of β and using the bound $\beta \leq \alpha + 1$. Therefore, we have that $X \in \mathcal{F}_1(\alpha)$ and the Shannon upper bound is achievable.

Note that the characteristic function of the optimal input distribution can be written as

$$\phi_X(t) = \frac{1 + 4t^2}{1 + 4\beta^2 t^2} = \frac{1}{\beta^2} + \left(1 - \frac{1}{\beta^2}\right) \frac{1}{1 + 4\beta^2 t^2} \quad (162)$$

$$= \frac{1}{\beta^2} + \left(1 - \frac{1}{\beta^2}\right) \phi_{X,1}(\beta t), \quad (163)$$

that is the optimal input distribution is a convex combination of the point mass at zero and the Laplace distribution. This concludes the proof.

APPENDIX L PROOF OF PROPOSITION 16

Show that

$$\lim_{\alpha \rightarrow \infty} \frac{C(\alpha; p, q)}{\frac{1}{p} \log(a_{p,q} \cdot \mathbb{E}[|X + Z|^p])} \leq 1. \quad (164)$$

follows trivially. Therefore, it remains to show that

$$\lim_{\alpha \rightarrow \infty} \frac{C(\alpha; p, q)}{\frac{1}{p} \log(a_{p,q} \cdot \mathbb{E}[|X + Z|^p])} \geq 1. \quad (165)$$

To that end we choose $X = X_D$ to be uniformly spaces equally distributed random variable with $N = \left\lfloor \left(\frac{p}{2} \mathbb{E}[|X|^p] + \frac{p}{2} \mathbb{E}[|Z|^p] \right)^{\frac{1}{p}} \right\rfloor$ and use the following bound from [15]

$$I(X_D; X_D + Z) \geq \log(N) - \text{gap}, \quad (166)$$

$$\text{gap} = G_r + \log \left(1 + \frac{\mathbb{E}^{\frac{1}{r}}[|X_D - f(Y)|^r]}{\mathbb{E}^{\frac{1}{r}}[|U|^r]} \right), \quad (167)$$

$$G_r = \log \left(\frac{\sqrt{\pi} (re)^{\frac{1}{r}} \Gamma(\frac{1}{r} + 1)}{\Gamma(\frac{1}{2} + 1) 2(r+1)^{\frac{1}{r}}} \right), \quad (168)$$

where $r \geq 1$, $f(\cdot)$ is any deterministic function and where U is uniform on $\left[-\frac{d_{\min}(X_D)}{2}, \frac{d_{\min}(X_D)}{2}\right]$. Moreover, we can further upper bound

$$\text{gap} \leq G_r + \log \left(1 + \frac{2(r+1)^{\frac{1}{r}} \mathbb{E}^{\frac{1}{r}}[|Z|^r]}{d_{\min}(X_D)} \right), \quad (169)$$

where the inequality follows by choosing $f(Y) = Y$ and observing that $\mathbb{E}[|U|^r] = \frac{d_{\min}^r(X_D)}{2^r(r+1)}$. Next, we show that for $\alpha \geq 1$ the gap is bound.

Note that

$$\frac{1}{d_{\min}^2(X_D)} = \frac{N^2 - 1}{12 \mathbb{E}[X_D^2]} \quad (170)$$

$$= \frac{\left(\lfloor \frac{p}{2} \mathbb{E}[|X_D|^p] + \frac{p}{2} \mathbb{E}[|Z|^p] \rfloor \right)^{\frac{2}{p}} - 1}{12 \mathbb{E}[X_D^2]} \quad (171)$$

$$\stackrel{a)}{\leq} \frac{\left(\frac{p}{2} \mathbb{E}[|X_D|^p] + \frac{p}{2} \mathbb{E}[|Z|^p] \right)^{\frac{2}{p}}}{12 \mathbb{E}[X_D^2]} \quad (172)$$

$$\stackrel{b)}{\leq} \frac{\left(\frac{p}{2} \mathbb{E}[|X_D|^p] + \frac{p}{2} \mathbb{E}[|Z|^p] \right)^{\frac{2}{p}}}{12 \mathbb{E}^{\frac{2}{p}}[|X_D|^p]} \quad (173)$$

$$= \frac{\left(\frac{p}{2} + \frac{\frac{p}{2} \mathbb{E}[|Z|^p]}{\mathbb{E}[|X_D|^p]} \right)^{\frac{2}{p}}}{12} \quad (174)$$

$$= \frac{\left(\frac{p}{2} + \frac{\mathbb{E}[|Z|^p]}{\alpha^p} \right)^{\frac{2}{p}}}{12} \quad (175)$$

$$\stackrel{c)}{\leq} \frac{\left(\frac{p}{2} + \mathbb{E}[|Z|^p] \right)^{\frac{2}{p}}}{12}, \quad (176)$$

where (in)-equality follow from: a) dropping the floor function; b) using the bound $\mathbb{E}[|X_D|^p] \leq 12^{\frac{p}{2}} \mathbb{E}^{\frac{p}{2}}[X^2]$ for any $p > 0$; and c) using that $\alpha \geq 1$.

Therefore, we have a bound

$$\text{gap} \leq G_1 + \log \left(1 + 2(r+1)^{\frac{1}{r}} \mathbb{E}^{\frac{1}{r}}[|Z|^r] \frac{\left(\frac{p}{2} + \mathbb{E}[|Z|^p] \right)^{\frac{1}{p}}}{\sqrt{12}} \right). \quad (177)$$

Next, observe that

$$I(X_D; X_D + Z) \geq \log(N) - \text{gap}, \quad (178)$$

$$= \log \left(\left[\left(\frac{p}{2} \mathbb{E}[|X_D|^p] + \frac{p}{2} \mathbb{E}[|Z|^p] \right)^{\frac{1}{p}} \right] \right) - \text{gap} \quad (179)$$

$$\stackrel{a)}{\geq} \frac{1}{p} \log \left(\frac{p}{2} \mathbb{E}[|X_D|^p] + \frac{p}{2} \mathbb{E}[|Z|^p] \right) + \log(2) - \text{gap} \quad (180)$$

$$\stackrel{b)}{\geq} \frac{1}{p} \log \left(\frac{p}{2} \mathbb{E}[|X + Z|^p] \right) + \frac{1}{p} \log \frac{p}{2 \max(2^{p-1}, 1)} + \frac{1}{p} \log(2) - \text{gap} \quad (181)$$

$$= \frac{1}{p} \log(a_{p,q} \mathbb{E}[|X_D + Z|^p]) + \frac{1}{p} \log \left(\frac{p}{2a_{p,q}} \right) + \log \frac{1}{\max(2^p, 1)} + \frac{1}{p} \log(2) - \text{gap}, \quad (182)$$

where inequalities follow from: a) $\lfloor x \rfloor \geq \frac{\lfloor x \rfloor}{2}$ for $x \geq 1$; and b) using the bound $\mathbb{E}[|X_D + Z|^p] \leq \max(2^{p-1}, 1) (\mathbb{E}[|X_D|^p] + \mathbb{E}[|Z|^p])$.

Therefore, we have that

$$\lim_{\alpha \rightarrow \infty} \frac{C(\alpha; p, q)}{\frac{1}{p} \log(a_{p,q} \cdot \mathbb{E}[|X_D + Z|^p])} \geq 1 - \lim_{\alpha \rightarrow \infty} \frac{\text{gap}}{\frac{1}{p} \log(a_{p,q} \cdot \mathbb{E}[|X_D + Z|^p])} = 1. \quad (183)$$

This concludes the proof.

APPENDIX M PROOF OF PROPOSITION 17

. The lower bound in (59) follows trivially. The upper bound follows by [31] where for any $X \in L^p$ and any X'

$$I(X; X + Z) \leq I(X'; X' + Z) + C_p, \quad (184)$$

$$C_p = \sup_{U \in \mathcal{F}_p(\alpha)} I(U; U + X'). \quad (185)$$

By taking $X' \sim \mathcal{N}_p(0, \alpha^p)$ and using Proposition 13 we have that

$$C_p = \sup_{U \in \mathcal{F}_p(\alpha)} I(U; U + X') \quad (186)$$

$$= \sup_{U \in \mathcal{F}_p(\alpha)} I(\alpha U; \alpha U + \alpha Z') \quad (187)$$

$$= \sup_{U \in \mathcal{F}_p(1)} I(U; U + Z') \quad (188)$$

$$= \frac{1}{p} \log(a_{p,p} \cdot \mathbb{E}[|U + Z'|^p]) \quad (189)$$

$$= \frac{1}{p} \log \left(\frac{p}{2} \cdot \mathbb{E}[|U + Z'|^p] \right). \quad (190)$$

Where for $p \leq 1$ we can use inequality $\mathbb{E}[|U + Z'|^p] \leq \mathbb{E}[|U|^p + |Z'|^p]$ to have that

$$C_p \leq \frac{1}{p} \log(2), \quad (191)$$

for $p \geq 1$ we can use inequality triangle inequality $\mathbb{E}^{\frac{1}{p}}[|U + Z'|^p] \leq \mathbb{E}^{\frac{1}{p}}[|U|^p] + \mathbb{E}^{\frac{1}{p}}[|Z'|^p]$ to have

$$C_p \leq \log(2); \quad (192)$$

Note that for a special case of $p = 2$ we have that $\mathbb{E}[|U + Z'|^2] = \mathbb{E}[|U|^2 + |Z'|^2]$ and hence

$$C_2 \leq \frac{1}{2} \log(2). \quad (193)$$

This concludes the proof.

APPENDIX N

PROOF OF THEOREM 20

A necessary and sufficient conditions from [26] states that a random variable X with distribution F is optimal if and only if

$$D(\mathcal{N}_q(x, 1) \| f_Y) = I(X; Y), x \in \mathcal{C}(F), \quad (194a)$$

$$D(\mathcal{N}_q(x, 1) \| f_Y) < I(X; Y), x \notin \mathcal{C}(F), \quad (194b)$$

where $\mathcal{C}(F)$ is a set of points of increase of F and f_Y is output pdf induced by F .

From (194) we have that for any $x_1 \in \mathcal{C}(F)$ and $x_2 \notin \mathcal{C}(F)$

$$0 < D(\mathcal{N}_q(x_1, 1) \| f_Y) - D(\mathcal{N}_q(x_2, 1) \| f_Y) = (x_1 - x_2) \frac{q}{2} \mathbb{E} \left[\mathbb{E} \left[Z^{q-1} | Y = \bar{Y} \right] \right], \quad (195)$$

where the last equality follow from Proposition 12. To check when $X = \pm\alpha$ is no longer optimal we have to check when is violated. Moreover, the next optimal distribution is ternary with $X = \{-A, 0, A\}$ Therefore, to check when $\pm\alpha$ equally likely is optimal choose $x_1 = \alpha$ and $x_2 = 0$ in (N). This concludes the proof.

REFERENCES

- [1] P. Lévy, *Calcul des probabilités*. Gauthier-Villars, 1925.
- [2] N. C. Beaulieu and D. J. Young, "Designing time-hopping ultrawide bandwidth receivers for multiuser interference environments," *Proceedings of the IEEE*, vol. 97, no. 2, pp. 255–284, 2009.
- [3] V. R. Algazi and R. M. Lerner, "Binary detection in white non-gaussian noise," *M.I.T. Lincoln Lab*, vol. 18, no. Res. DS-2138, pp. 241–250.
- [4] J. Miller and J. Thomas, "Detectors for discrete-time signals in non-gaussian noise," *IEEE Transactions on Information Theory*, vol. 18, no. 2, pp. 241–250, 1972.
- [5] S. Banerjee and M. Agrawal, "Underwater acoustic noise with generalized Gaussian statistics: Effects on error performance," in *OCEANS - Bergen, 2013 MTS/IEEE*, June 2013, pp. 1–8.
- [6] O. Bernard, J. D'hooge, and D. Fribouler, "Statistical modeling of the radio-frequency signal in echocardiographic images based on generalized gaussian distribution," in *3rd IEEE International Symposium on Biomedical Imaging: Nano to Macro, 2006*. IEEE, 2006, pp. 153–156.
- [7] D. Gonzalez-Jimenez, F. Perez-Gonzalez, P. Comesana-Alfaro, L. Perez-Freire, and J. L. Alba-Castro, "Modeling gabor coefficients via generalized gaussian distributions for face recognition," in *2007 IEEE International Conference on Image Processing*, vol. 4, Sept 2007, pp. IV – 485–IV – 488.
- [8] M. O. M. Mahmoud, M. Jaidane-Saidane, J. Souissi, and N. Hizaoui, "Modeling of the load duration curve using the asymmetric generalized gaussian distribution: Case of the tunisian power system," in *Power and Energy Society General Meeting-Conversion and Delivery of Electrical Energy in the 21st Century, 2008 IEEE*. IEEE, 2008, pp. 1–7.
- [9] M. S. Davis, P. Bidigare, and D. Chang, "Statistical modeling and ml parameter estimation of complex sar imagery," in *2007 Conference Record of the Forty-First Asilomar Conference on Signals, Systems and Computers*, Nov 2007, pp. 500–502.
- [10] M. N. Do and M. Vetterli, "Wavelet-based texture retrieval using generalized gaussian density and kullback-leibler distance," *IEEE Transactions on Image Processing*, vol. 11, no. 2, pp. 146–158, Feb 2002.
- [11] S. Nadarajah, "A generalized normal distribution," *Journal of Applied Statistics*, vol. 32, no. 7, pp. 685–694, 2005.
- [12] H. Poor and J. Thomas, "Locally optimum detection of discrete-time stochastic signals in non-gaussian noise," *The Journal of the Acoustical Society of America*, vol. 63, no. 1, pp. 75–80, 1978.
- [13] T. Cover and J. Thomas, *Elements of Information Theory: Second Edition*. Wiley, 2006.
- [14] E. Lutwak, D. Yang, and G. Zhang, "Moment-entropy inequalities for a random vector," *IEEE Trans. Inf. Theory*, vol. 53, no. 4, pp. 1603–1607, April 2007.
- [15] A. Dytso, R. Bustin, D. Tuninetti, N. Devroye, S. Shamai, and H. V. Poor, "On the minimum mean p -th error in Gaussian noise channels and its applications." Submitted to *IEEE Trans. Inf. Theory*, <https://arxiv.org/abs/1607.01461>, 2016.
- [16] L. Ozarow and A. Wyner, "On the capacity of the Gaussian channel with a finite number of input levels," *IEEE Trans. Inf. Theory*, vol. 36, no. 6, pp. 1426–1428, Nov 1990.
- [17] F. Nielsen and R. Nock, "A series of maximum entropy upper bounds of the differential entropy," 2016.
- [18] T. K. Pogány and S. Nadarajah, "On the characteristic function of the generalized normal distribution," *Comptes rendus mathématique*, vol. 348, no. 3, pp. 203–206, 2010.
- [19] H. Soury and M.-S. Alouini, "New results on the sum of two generalized gaussian random variables," in *2015 IEEE Global Conference on Signal and Information Processing (GlobalSIP)*. IEEE, 2015, pp. 1017–1021.
- [20] R. Vasudeva and J. V. Kumari, "On general error distributions," 2013.
- [21] A. Dytso, R. Bustin, D. Tuninetti, N. Devroye, H. V. Poor, and S. Shamai, "On the applications of the minimum mean p -th error (mmpe) to information theoretic quantities," in *2016 IEEE Information Theory Workshop (ITW)*, Sept 2016, pp. 66–70.
- [22] D. Guo, S. Shamai, and S. Verdú, "Mutual information and minimum mean-square error in Gaussian channels," *IEEE Trans. Inf. Theory*, vol. 51, no. 4, pp. 1261–1282, April 2005.
- [23] —, "Additive non-gaussian noise channels: Mutual information and conditional mean estimation," in *Proceedings. International Symposium on Information Theory, 2005. ISIT 2005*. IEEE, 2005, pp. 719–723.
- [24] S. Verdú, "The exponential distribution in information theory," *Problemy peredachi informatsii*, vol. 32, no. 1, pp. 100–111, 1996.
- [25] J. Fahn and I. Abou-Faycal, "Input constraints and noise density functions: A simple relation for bounded-support and discrete capacity-achieving inputs," *arXiv preprint arXiv:1602.00878*, 2016.
- [26] J. G. Smith, "The information capacity of amplitude-and variance-constrained scalar Gaussian channels," *Information and Control*, vol. 18, no. 3, pp. 203–219, 1971.
- [27] S. Bochner *et al.*, "Stable laws of probability and completely monotone functions," *Duke Mathematical Journal*, vol. 3, no. 4, pp. 726–728, 1937.
- [28] N. Elkies, A. Odlyzko, and J. Rush, "On the packing densities of superballs and other bodies," *Inventiones mathematicae*, vol. 105, no. 1, pp. 613–639, 1991.
- [29] N. G. Ushakov, *Selected topics in characteristic functions*. Walter de Gruyter, 1999.
- [30] B. Brown, "Formulae for absolute moments," *Journal of the Australian Mathematical Society*, vol. 13, no. 01, pp. 104–106, 1972.
- [31] R. Zamir and U. Erez, "A Gaussian input is not too bad," *IEEE Trans. Inf. Theory*, vol. 50, no. 6, pp. 1362–1367, June 2004.